

Alyssa Hee

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EDUCATION

University of Toronto

Bachelor of Applied Science in Electrical and Computer Engineering + PEY Co-op

Sept 2024 – Apr 2028

CGPA: 3.77/4.00

- Intended Minor in Sustainable Energy

EXPERIENCE

Undergraduate Research Assistant

May 2026 – Present

University of Manitoba RIGA Lab

- Developing a Dual Active Bridge (DAB) converter running at 200kHz for use in a Solid State Transformer
- Designed and validated the converter in PLECS, implementing phase-shift modulation and closed-loop voltage control while analyzing efficiency and soft-switching performance
- Building a physical prototype for controller Hardware-in-the-Loop testing using a PLECS RT Box and TI C2000

Technical Team Co-Director

May 2026 – Present

University of Toronto IEEE Student Branch

Toronto, ON

- Develop and deliver summer tutorials covering PCB design, from circuit simulation and schematic to routing and testing using LTSpice and Altium Designer
- Planned and led an introductory workshop on Arduino, coordinating content and team collaboration to deliver an engaging hands-on learning experience
- Co-lead an Altium Designer workshop focused on schematic, PCB layout, and routing fundamentals

Co-Webmaster

July 2025 – Aug 2025

F!rosh Week

- Collaborated with a team to redesign and maintain *orientation.skule.ca* using React, CSS, and MongoDB, improving responsiveness and supporting 1,000+ registrants
- Deployed and maintained site updates through Git and SSH, troubleshooting user accounts and technical issues

SELECTED PROJECTS

Sparky PCB

June 2026 – Present

- Designed and simulated a battery-powered electrostatic discharge (ESD) detection circuit using a 555 timer, MOSFET switching network, and TVS protection to detect positive and negative ESD events
- Created a 2-layer PCB in Altium Designer, optimizing ESD current return paths with ground planes, stitching vias, decoupling capacitors, and surge protection layout techniques
- Validated circuit operation from ± 1 to ± 35 kV in LTSpice with IEC 61000-4-2 and Human Body Model ESD simulations

FPGA Magic Flute

Mar 2026 – Apr 2026

- Built an FPGA-based digital wind instrument with custom 3D-printed flute keys designed in Fusion360
- Integrated microphone and flute keys with Nios V processor using GPIO ports for real-time instrument control
- Implemented a finite-state machine in C to control audio sample playback and microphone-based breath detection across multiple playable sound modes

TECHNICAL SKILLS

CAD/Modelling: PLECS, LTSpice, Altium Designer, Fusion 360, Blender

Hardware: Circuit prototyping, soldering, PCB debugging

Programming: C, C++, Verilog, JavaScript, HTML/CSS Swift, SwiftUI, React

Developer Tools: Git, VS Code, Figma, Pixilart